ADVANCED SHORT CIRCUIT PREVENTIVE MECHANISM FOR ELECTRIC VEHICLES

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report "**ADVANCED SHORT CIRCUIT PREVENTIVE MECHANISM FOR ELECTRIC VEHICLES**" is the bonafide work of **"SHANJAY RK (927621BEC195), SIVARAMAKRISHNAN M (927621BEC201), SUBASH P (927621BEC213), YASAR S (927621BEC245)"** who carried out the project work under my supervision in the academic year 2024-2025.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

INSTITUTION VISION AND MISSION

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To emerge as a leader among the top institutions in the field of technical education.

Mission

M1: Produce smart technocrats with empirical knowledge who can surmount the global challenges.

M2: Create a diverse, fully -engaged, learner -centric campus environment to provide quality education to the students.

M3: Maintain mutually beneficial partnerships with our alumni, industry and professional associations

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M3: Provide entrepreneurial skills and leadership qualities.

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PEO1: **Core Competence:** Graduates will have a successful career in academia or industry associated with Electronics and Communication Engineering

PEO2: **Professionalism:** Graduates will provide feasible solutions for the challenging problems through comprehensive research and innovation in the allied areas of Electronics and Communication Engineering.

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Program Outcomes

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

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PSO2: Able to solve complex problems in Electronics and Communication Engineering with analytical and managerial skills either independently or in team using latest hardware and software tools to fulfil the industrial expectations.

Abstract	Matching with POs, PSOs
RNN , LSTM , EV, BMS , AI , IOT , UART, PWM	PO1, PO2, PO6, PO7, PO9, PO12, PSO1, PSO2

ACKNOWLEDGEMENT

We gratefully remember our beloved **Founder Chairman, (Late) Thiru. M. Kumarasamy**, whose vision and legacy laid the foundation for our education and inspired us to successfully complete this project.

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ABSTRACT

In order to improve safety, dependability, and system performance, the proposed project introduces a cutting- edge short circuit preventative mechanism made especially for electric vehicles (EVs). In EV power systems, short circuits are a serious problem since they can result in serious harm, component failures, and safety risks. Traditional protection techniques, including circuit breakers and fuses, are frequently slow, reactive, and unable to handle the complex and dynamic working circumstances of contemporary EVs to proactively prevent short circuits, this research makes use of a combination of high-speed sensing, predictive fault detection algorithms, and intelligent control tactics. The system uses machine learning techniques to predict and alleviate issues before they increase by continuously monitoring critical metrics including current, voltage, and temperature variations. To guarantee quick reaction and strong protection, sophisticated parts are integrated, such as semiconductor switches and thermal management subsystems, the suggested mechanism also includes self-diagnostic capabilities, which allow for continuous monitoring and automatic recovery in the event of temporary defects. The system's ability to detect and prevent short circuits is demonstrated by simulation and experimental findings, which beat conventional protection techniques in terms of speed, accuracy, and dependability This invention contributes to safer, more effective, and sustainable EV systems by offering a comprehensive solution to one of the most important problems in electric mobility. By laying the groundwork for next-generation electrical safety solutions, the project hopes to increase EV adoption and confidence.

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
AI	Artificial Intelligence
BMS	Battery Management System
CPU	Central Processing Unit
IoT	Internet of Things
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LSTM	Long Short-Term Memory
ML	Machine Learning
PWM	Pulse Width Modulation
RAM	Random Access Memory
RNN	Recurrent Neural Network
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver Transmitter
USB	Universal Serial Bus
VDC	Volts Direct Current

CHAPTER 1

INTRODUCTION

1.1 Objective

The growing popularity of electric vehicles (EVs) has brought attention to how important it is to have dependable and effective electrical systems in order to guarantee both safety and peak performance. Short circuits are one of the many issues that EV power systems encounter, and they are especially dangerous since they can result in serious harm, such as system failures, thermal runaway, and even user safety risks. As a result, a top focus in automobile research and development is now creating sophisticated preventive mechanisms to identify, lessen, and control short circuits in EV power systems. Traditional short circuit protection techniques frequently involve fuses or circuit breakers, which have intrinsic drawbacks such slower response times, no real-time monitoring, and restricted reusability, despite being useful in some situations.

These difficulties call for the creation of increasingly complex systems that can detect faults proactively, react quickly, and recover on their own. This study suggests a sophisticated short circuit prevention system made especially for electric cars that incorporates state-of-the-art technology including active power flow management, machine learning- based predictive diagnostics, and real-time fault detection algorithms. The risk of short circuits is further increased by the dynamic and constantly shifting working circumstances of EV power systems, which include fluctuating loads, high currents, and extremely high temperatures.

1.2 Significance of the Project

In order to provide thorough and dependable protection for the entire system, effective short circuit avoidance systems must take this complexity into consideration by including features like adaptive control techniques, fault-tolerant designs, and improved thermal management. The proposed mechanism not only enhances the safety and reliability of EV power systems but also contributes to extending the lifespan of critical components and reducing downtime. By integrating advanced analytics and control strategies, this solution aligns with the broader goal of fostering sustainable and safe transportation technologies. The effectiveness of the mechanism is validated through simulation and experimental results, demonstrating significant improvements in fault prevention and system robustness compared to the effective and traditional approaches.

Electric vehicles (EVs) are rapidly gaining popularity due to their potential to reduce greenhouse gas emissions and dependence on fossil fuels but there are still significant issues with EV power systems' dependability and security. Short circuits, which can have disastrous consequences like component failure, thermal runaway, fire dangers, and even the risk of human death, are a major worry in these systems. A number of things can cause short circuits, such as poor insulation, conductive objects, bad wiring, or unforeseen environmental circumstances like moisture or extremely high or low temperatures.

1.3 Emergence of Smart Protection Systems

EVs are susceptible to fast cycles of charging and discharging and operate under a variety of load circumstances, which can put stress on the electrical system. Furthermore, traditional systems are reactive rather than proactive in their approach since they are unable to monitor, predict, and avoid failures in real-time. The development of more effective and efficient protective methods has been made possible in recent years by developments in semiconductor technology, sensor systems, and intelligent control units. High-speed switches, advanced algorithms, and machine learning tools enable real-time monitoring and predictive fault detection, paving the way for active and preemptive short circuit prevention.

The hardware implementation of the Advanced Short Circuit Preventive Mechanism for Electric Vehicles involves a combination of crucial components, including sensors, microcontrollers, and control systems, all working in unison to ensure the safety and optimal performance of the vehicle's electrical system. At the heart of the system lies the microcontroller (PIC), which acts as the central processing unit. The microcontroller collects data from various sensors, such as current, temperature, and voltage sensors, and processes this information to detect potential issues like short circuits or abnormal cell balancing in the battery. The current sensor plays a vital role in monitoring the current flowing through the vehicle's power system.

CHAPTER 2

LITERATURE REVIEW

2.1 Design and Evaluation of High-Speed Overcurrent and Short-Circuit Detection Circuits With High Noise Margin for WBG* Power Semiconductor Devices

H.-C Park et al. (2024) introduced a novel diagnostic approach for identifying internal short circuits in lithium-ion batteries by leveraging relaxation voltage measurements. Their proposed method presents a significant advancement in quantitative fault detection, providing a more reliable and scientific basis for assessing battery health and integrity. The study emphasizes the critical role of real-time monitoring in enhancing the overall safety and reliability of electric vehicles (EVs), especially under varying load and operational conditions.

One of the key contributions of their work is the demonstration that relaxation voltage profiles can be used to detect internal faults at an early stage, well before they evolve into severe or catastrophic failures. This proactive fault detection strategy aligns well with modern trends in EV protection systems, which are increasingly shifting from reactive to predictive models.

Despite the promise of this technique, the authors also pointed out a crucial limitation: integration challenges. Specifically, embedding this diagnostic method into existing EV battery management systems may introduce latency issues, potentially affecting the speed of fault response.

2.2 Inter-Turn Short Circuit Faults and Mitigation Measures for Dual-Three Phase Machines

D. Walch et al. (2024) presented an online, data-driven fault diagnosis system combined with thermal runaway early warning for electric vehicle batteries. They employed machine learning algorithms to predict potential faults, enabling the system to detect early signs of thermal events before they escalate. While the system significantly improved safety, the authors emphasized that the high computational cost and real-time processing requirements could limit its practical application in some cases. Introduced a digital gate driver IC that features digitally adjustable DESAT (desaturation detection) to prevent false detection and improve short-circuit protection. This work is relevant to EV systems as it enhances the protection of power semiconductors, a critical component in EV powertrains. However, the integration of such advanced gate driver ICs in commercial EVs needs further development in terms of reliability and cost-effectiveness.

2.3 Electric Drive Technology Trends, Challenges, and Opportunities for Future Electric Vehicles

I. Hussain et al. (2021) presented a Lithium-ion batteries are widely used in electric vehicles and portable electronics due to their high energy density, long cycle life, and efficiency. However, safety concerns remain a major challenge, especially when external short circuits occur. An external short circuit happens when a low-resistance connection is unintentionally established between the battery terminals, allowing a large current to flow. This can lead to rapid heating, thermal runaway, fire, or even explosion, posing serious risks to users and surrounding equipment.

2.4 Quantitative Diagnosis of Internal Short Circuit for Lithium-Ion Batteries Using Relaxation Voltage

The study of external short circuit risks involves analysing the thermal and electrical behaviour of battery modules under fault conditions. Factors such as the internal resistance of the cells, current path, ambient temperature, and duration of the short circuit play critical roles in determining the severity of the damage. Advanced simulation tools and real-world testing are used to evaluate how different battery chemistries and configurations respond to such events.

2.5 An Online Data-Driven Fault Diagnosis and Thermal Runaway Early Warning for Electric Vehicle Batteries

To mitigate these risks, effective protection designs are crucial. These include integrating fast-acting fuses, current limiters, and circuit breakers within the battery pack. Battery Management Systems (BMS) also monitor voltage, current, and temperature in real-time and can disconnect the battery during abnormal conditions. In conclusion, understanding and addressing external short circuit risks is vital for enhancing the safety and reliability of lithium-ion battery systems. With proper design, testing, and protective strategies, the likelihood and impact of short circuits can be significantly reduced, promoting safer usage in EVs and other high-demand applications. With the adoption of comprehensive safety protocols—including proper system design, rigorous testing, and intelligent monitoring—the likelihood and consequences of short circuits can be substantially minimized, ensuring safer performance in electric vehicles and other high-demand energy applications.

Table 2.1 Literature Survey

Author	Title	Publication Details	Remarks
H.-C. Park et al.	"Design and Evaluation of High-Speed Overcurrent and Short-Circuit Detection Circuits With High Noise Margin for WBG* Power Semiconductor Devices"	2024, IEEE Access	Focuses on device-level protection circuits; lacks application to large-scale EV systems or grid-connected setups.
D. Walch et al.	"Inter-Turn Short Circuit Faults and Mitigation Measures for Dual-Three Phase Machines"	2024, ICEM Conference	Limited to specific machine configurations; applicability to other motor types or EV systems not explored.
I. Husain et al.	"Electric Drive Technology Trends, Challenges, and Opportunities for Future Electric Vehicles"	2021, Proceedings of the IEEE	Broad overview of trends; does not provide deep technical insight into fault detection or mitigation strategies.
D. Qiao et al.	"Quantitative Diagnosis of Internal Short Circuit for Lithium-Ion Batteries Using Relaxation Voltage"	2024, IEEE Transactions on Industrial Electronics	Focuses primarily on diagnosis without providing real-time mitigation strategies for detected faults.
Z. Sun et al.	"An Online Data-Driven Fault Diagnosis and Thermal Runaway Early Warning for Electric Vehicle Batteries"	2022, IEEE Transactions on Power Electronics	Limited to battery systems; lacks consideration of external factors like environmental conditions.

CHAPTER 3

EXISTING SYSTEM

3.1 Conventional method

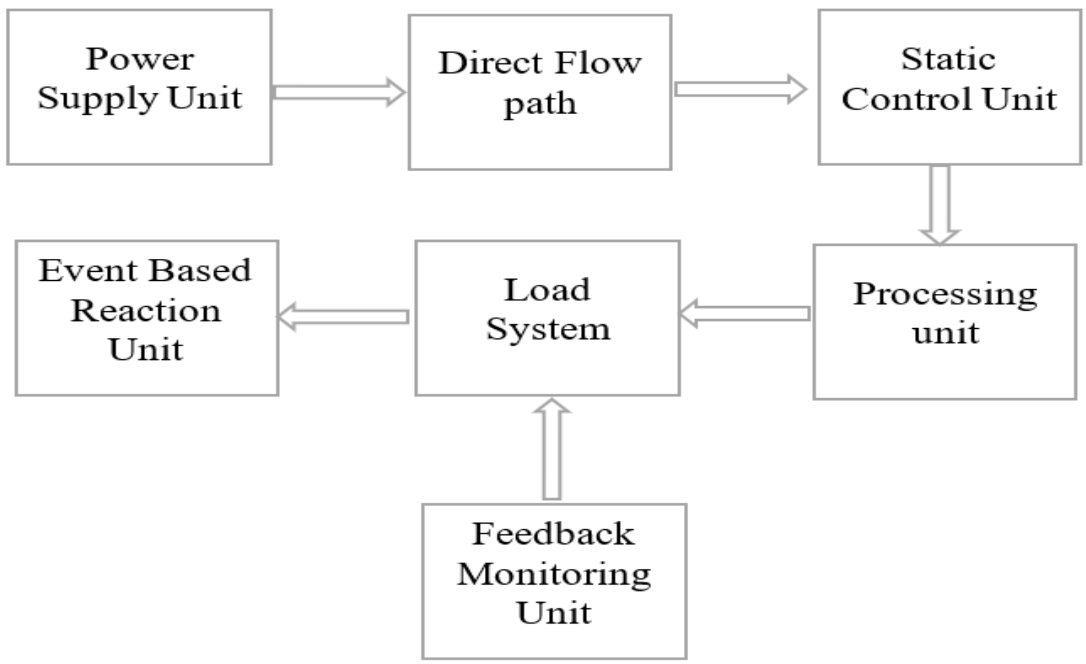

Fig 3.1 Block diagram of Existing method

Electric vehicle (EV) safety, especially in terms of battery protection and fault detection, continues to be a critical issue. Existing systems primarily rely on traditional methods such as fuses and circuit breakers for short circuit protection. While effective to some extent, these methods are reactive rather than proactive, only responding after a fault has already occurred. As a result, they cannot prevent damage to the vehicle or its components, often limiting the overall effectiveness of the safety system. These sensors tend to operate in isolation without being integrated into a centralized system that can analyse data from multiple sources simultaneously.

The current fault detection techniques used in EVs are often based on basic overcurrent protection and temperature monitoring. These systems rely on set thresholds to identify when the current exceeds a safe limit, which can signal a short circuit or overcharge. While these systems may detect faults once they have occurred, they do not have the ability to predict faults in advance or take proactive measures before damage happens.

Similarly, temperature monitoring can detect overheating, but it is only effective after the temperature has risen, which again limits its ability to act in time to prevent significant damage. Moreover, the existing systems lack predictive capabilities. They do not integrate advanced analytics, machine learning, or artificial intelligence to assess the health of the vehicle's battery or power system based on real-time data. This results in missed opportunities for early fault detection, such as detecting cell imbalance or thermal runaway before they escalate. Predictive analytics could help in identifying issues well before they become critical, but it is not commonly incorporated into current EV safety systems.

Current monitoring and diagnostic systems are often manual, requiring periodic checks or human intervention to assess the state of the vehicle's electrical components. While diagnostic tools have improved, they still do not offer continuous, real-time data analysis that could automatically detect faults or prevent issues before they arise. These manual systems are time-consuming and prone to human error, leading to delayed responses and reduced effectiveness in preventing vehicle downtime or costly repairs. Additionally, most existing systems rely on individual sensors to monitor faults like voltage drop, current anomalies, or temperature fluctuations.

3.2 Drawbacks of Existing System

The disadvantage of the existing systems in EVs rely on reactive approaches to fault detection and do not leverage advanced technologies like machine learning or real-time data analytics. These limitations underscore the need for a more intelligent, proactive safety system capable of predicting and preventing faults before they result in significant damage, which is the driving force behind the proposed project. The microcontroller collects data from various sensors, such as current, temperature, and voltage sensors, and processes this information to detect potential issues like short circuits or abnormal cell balancing in the battery.

The current sensor plays a vital role in monitoring the current flowing through the vehicle's power system he existing EV systems face several drawbacks such as limited charging infrastructure, long charging times, and poor battery health monitoring, which affect user convenience and reliability. Inaccurate range prediction and lack of smart load management can lead to inefficient energy use and potential grid overload. However, these sensors tend to operate in isolation without being integrated into a centralized system that can analyse data from multiple sources simultaneously. This lack of integration means that the system may fail to correlate data from various sensors, which could lead to missed faults or delayed responses to developing issues.

CHAPTER 4

PROPOSED SYSTEM

4.1 System Overview

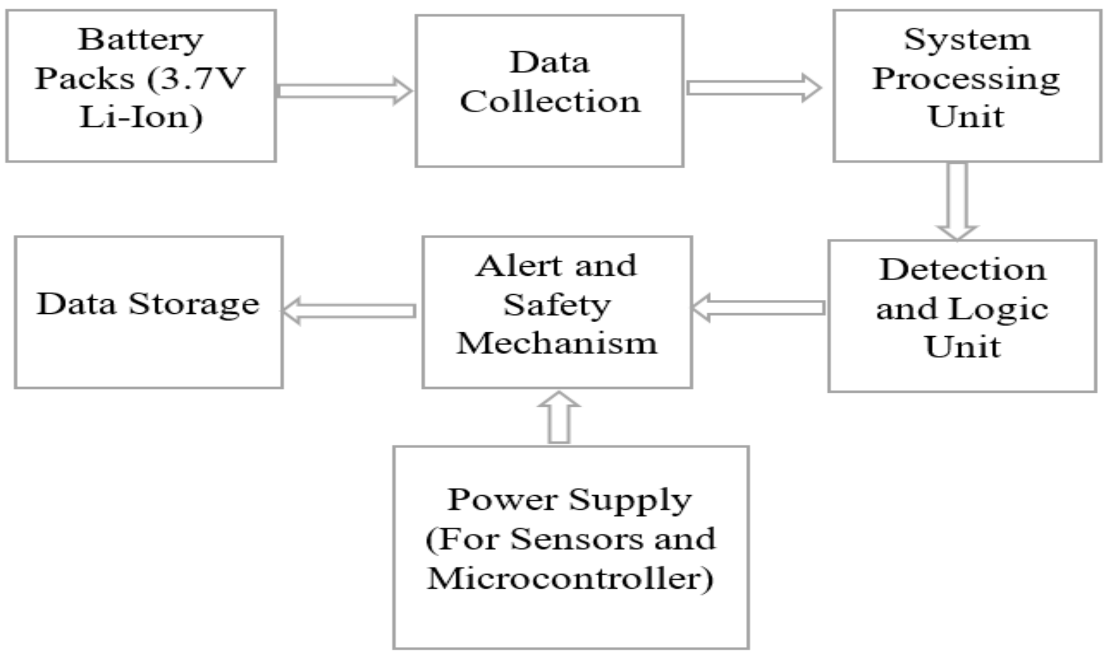

Fig 4.1 Block diagram of Proposed method

The proposed system for detecting and preventing short circuits in electric vehicles utilizes an advanced combination of hardware components and a well-defined process flow, ensuring the safety and longevity of the vehicle's battery packs. The system begins with the two cylindrical Li-ion cells, which power the entire system. These cells provide the necessary energy for sensors, the microcontroller, and other system components, ensuring continuous operation.

Data collection plays a pivotal role in the system's function. It involves three main sensors: the ACS712 current sensor, ADS1115 cell voltage monitor, and DS18B20 temperature sensor. The ACS712 current sensor measures the current flowing through the battery packs, allowing the system to detect any abnormal currents indicative of potential short circuits or overcurrent situations. Simultaneously, the ADS1115 monitors the voltage of each individual cell in the battery pack. This is essential for detecting any imbalances between cells, which could signal a fault. The DS18B20 temperature sensor keeps track of the battery temperature, as excessive heat can be a sign of a potential short circuit or other malfunction.

The heart of the system lies in the PIC16F877A Microcontroller. This microcontroller processes the data collected from the sensors and runs logical algorithms to determine if the battery pack is in a potentially dangerous state. Once an issue, such as an imbalance or abnormal current, is detected, the system immediately triggers its alert and safety mechanisms. The Relay Module serves as a crucial component in this safety system; it disconnects the battery pack from the system in case of a detected hazard, preventing further damage or potential fires. To provide clear feedback to the user, LED Indicators are employed to show the operational status of the system, such as normal, warning, or critical conditions, offering immediate visual alerts. For remote monitoring and control, the ESP8266 Wi-Fi module enables the system to connect to a wireless network, allowing users to track the vehicle's status and receive alerts through a mobile app or web interface. This integration ensures that the vehicle owner can monitor the system even when they are not in direct proximity to the vehicle .

Finally, all the data collected from the sensors are logged into a storage system. This data is useful for diagnostics, detecting recurring patterns, and performing any required maintenance or analysis over time. In conclusion, the proposed system integrates a comprehensive suite of components, including the microcontroller, sensors, safety mechanisms, and remote monitoring capabilities, to ensure the safe operation of electric vehicle battery packs. By continuously monitoring critical parameters such as current, voltage, and temperature, the system is capable of detecting potential faults in real-time, taking proactive safety measures, and alerting the user, thus preventing damage and ensuring the longevity of the battery system.

4.2 Machine Learning Approach

Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN), play a pivotal role in detecting and preventing short circuits and other faults in the electric vehicle's battery packs. LSTM networks are particularly well-suited for this application because they excel at handling time-series data, which is crucial for continuous monitoring of battery parameters such as current, voltage, and temperature. These parameters change over time, and understanding their temporal relationships is essential for accurate fault prediction. This is especially important because faults in the battery pack, such as short circuits, voltage imbalances, or overheating, often develop gradually, manifesting as subtle changes over time that would be hard to detect using conventional methods.

4.3 Role of LSTM

The system uses LSTM networks to analyse the sensor data, which includes real-time measurements of current, voltage, and temperature. Unlike traditional machine learning algorithms that operate on static data, LSTM networks are designed to retain information over time, allowing the system to not only consider the current sensor readings but also how these readings have evolved. This is especially important because faults in the battery pack, such as short circuits, voltage imbalances, or overheating, often develop gradually, manifesting as subtle changes over time that would be hard to detect using conventional methods. For example, in a battery pack, a gradual increase in temperature or a slow change in current might indicate an impending short circuit or thermal runaway, but these changes may not be severe enough to trigger a traditional threshold-based safety mechanism immediately.

LSTM networks can learn these patterns over time and identify trends that precede fault conditions. By doing so, the system can predict future states and take proactive measures before the problem escalates, ensuring the safety of the vehicle. Furthermore, the LSTM model’s ability to handle complex time-series data allows it to adapt to varying operational conditions, making it a reliable solution for long-term monitoring of electric vehicle systems. such as the relay module to disconnect the battery pack or the cooling system to prevent overheating. This real-time fault detection ensures that the system can respond quickly and effectively to potential risks, minimizing the chances of a catastrophic failure.

4.4 Predictive Maintenance with LSTM

The LSTM model is trained using historical data from the sensors, where normal operating conditions are compared with data collected during fault scenarios. Through this training, the LSTM network learns to identify patterns and sequences in the data that are indicative of potential failures.

Once trained, the LSTM model can predict the likelihood of a fault based on the real-time sensor data, providing early warnings even when the parameters appear to be within normal limits. For instance, the LSTM network could be used to predict the temperature trend of the battery pack. If the temperature is steadily rising, the system might flag this as a potential problem, even if the temperature is still below critical thresholds. The system could then activate the cooling system or send an alert to the user to take preventative action, preventing further damage.One of the key advantages of LSTM networks is their ability to detect anomalies and classify faults based on temporal data.

The system can be trained to recognize specific fault signatures, such as irregular spikes in current or voltage, which could indicate issues like short circuits or internal damage to the battery cells. LSTM networks can also differentiate between types of faults based on how these parameters change over time. For example, a short circuit might cause an immediate surge in current, whereas an imbalanced battery pack might lead to slower, gradual changes in voltage. By continuously monitoring the sensor data, the LSTM model can flag any deviations from the learned patterns, triggering an alert or activating the safety mechanisms.

4.5 Hardware Implementation

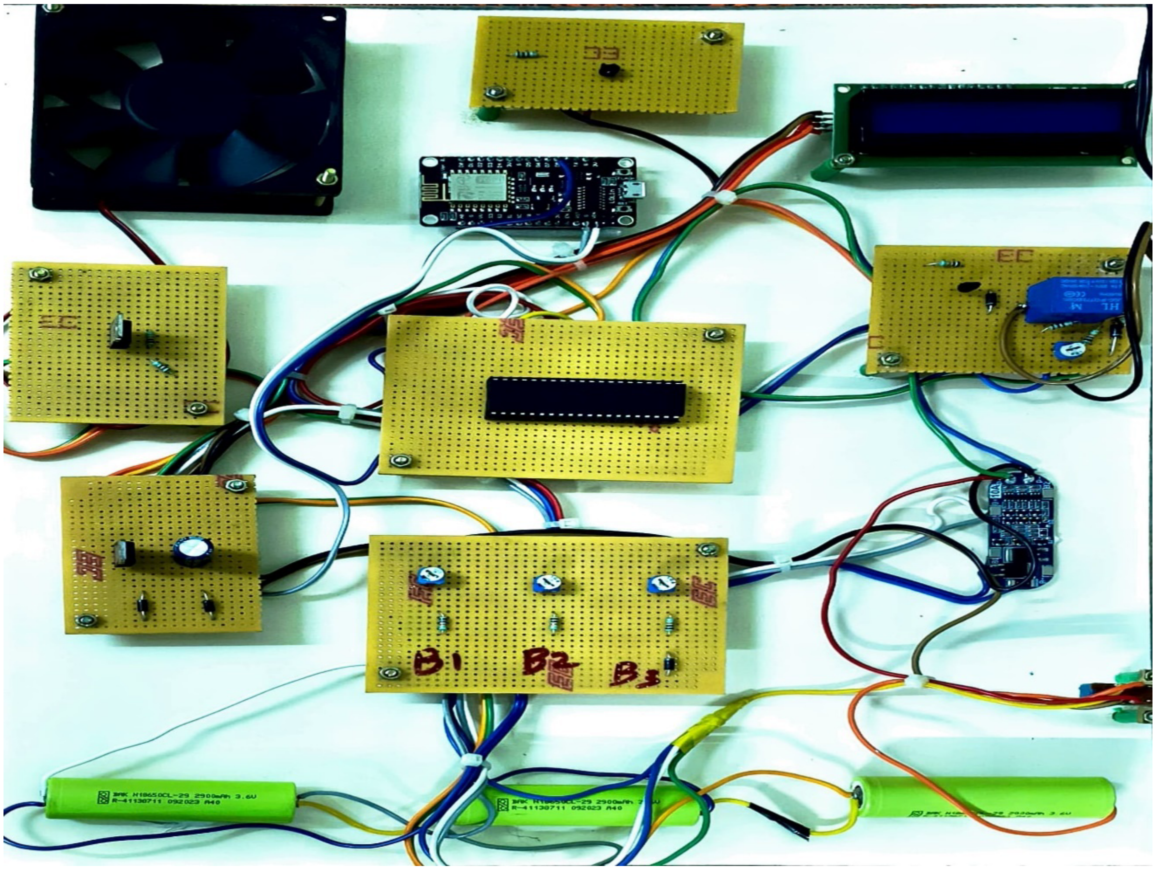

Fig.4.2 Assembled Hardware

The hardware implementation of the Advanced Short Circuit Preventive Mechanism for Electric Vehicles involves a combination of crucial components, including sensors, microcontrollers, and control systems, all working in unison to ensure the safety and optimal performance of the vehicle's electrical system. At the heart of the system lies the microcontroller (PIC), which acts as the central processing unit. The microcontroller collects data from various sensors, such as current, temperature, and voltage sensors, and processes this information to detect potential issues like short circuits or abnormal cell balancing in the battery.

It provides real-time feedback to the microcontroller, allowing it to identify sudden spikes in current that could indicate a short circuit or electrical fault. Similarly, the temperature sensor continuously monitors the temperature of critical components such as the battery. If the temperature exceeds safe thresholds, it could be a sign of overheating, often caused by a short circuit, and the system will react accordingly to prevent damage. The voltage sensor measures the voltage levels of the battery, ensuring that the voltage remains within safe limits. Any significant fluctuation in voltage may point to a potential fault, and the system can take immediate action to mitigate the risk.

In addition to these sensors, the system is equipped with a cell balancing circuit, which ensures that all cells within the battery pack maintain the same voltage level, thus optimizing the battery's overall efficiency and longevity. If the cells become unbalanced, the system can take corrective action to restore balance, preventing further issues. The relay or control mechanism is another key component that disconnects the battery or specific parts of the vehicle's electrical system when an anomaly is detected. This relay helps isolate the problem, preventing further damage and protecting the vehicle's components. For remote monitoring and alerts, the system can incorporate a communication module, such as Wi-Fi or Bluetooth, allowing real-time data transmission to mobile devices or cloud platforms. This enables the vehicle owner or technician to monitor the vehicle's health remotely, receive alerts in case of faults, and make timely interventions if necessary. The power supply unit is responsible for providing stable power to the microcontroller and all connected sensors, ensuring that the system operates without interruption.

By integrating advanced sensors and control mechanisms, the system can enhance the safety and reliability of electric vehicles, ensuring early detection and prevention of short circuits and other electrical faults. The seamless integration of these hardware components is vital for ensuring the effective operation of the vehicle and providing peace of mind to users regarding their vehicle's safety. The system also includes overvoltage and overcurrent protection circuits, which act as an additional layer of safety. These circuits monitor the system for any extreme voltage or current conditions that could damage the components or lead to dangerous situations.

If any threshold is exceeded, these protection mechanisms automatically disconnect the affected parts of the system, preventing further damage and protecting the vehicle's overall electrical infrastructure. The hardware implementation of the system successfully integrates essential components to ensure reliable performance, safety, and efficiency. Through careful selection and configuration of sensors, controllers, and protection circuits, the design meets the functional requirements of the project. The system demonstrates stable operation under real-time conditions, validating the effectiveness of the hardware architecture. Additionally, proper layout, testing, and troubleshooting have helped minimize errors and optimize the overall setup. This implementation lays a strong foundation for further enhancements and real-world deployment of the project.

CHAPTER 5

COMPONENTS USED

5.1 PIC Microcontroller

The original PIC was built to be used with General Instrument's new CP1600 16-bit CPU. While generally a good CPU, the CP1600 had poor I/O performance, and the 8-bit PIC was developed in 1975 to improve performance of the overall system by offloading I/O tasks from the CPU. The PIC used simple microcode stored in ROM to perform its tasks, and although the term was not used at the time, it shares some common features with RISC designs. In 1985, General Instrument spun off their microelectronics division and the new ownership cancelled almost everything which by this time was mostly out-of-date.

The PIC, however, was upgraded with an internal EPROM to produce a programmable channel controller. Today a huge variety of PICs are available with various on-board peripherals (serial communication modules, UARTs, motor control kernels, etc.) and program memory from 256 words to 64k words and more (a "word" is one assembly language instruction, varying from 8, 12, 14 or 16 bits depending on the specific PIC micro family).PIC and PIC micro are registered trademarks of Microchip Technology. It is generally thought that PIC stands for Peripheral Interface Controller, although General Instruments' original acronym for the initial PIC1640 and PIC1650 devices was "Programmable Interface Controller". The acronym was quickly replaced with "Programmable Intelligent Computer".

The Microchip 16C84 (PIC16x84), introduced in 1993, was the first[citation needed] Microchip CPU with on-chip EEPROM memory. This electrically erasable memory made it cost less than CPUs that required a quartz "erase window" for erasing EPROM. By 2013, Microchip was shipping over one billion PIC microcontrollers every year. The architectural decisions are directed at the maximization of speed-to-cost ratio. The PIC architecture was among the first scalar CPU designs, and is still among the simplest and cheapest. The Harvard architecture—in which instructions and data come from separate sources—simplifies timing and microcircuit design greatly, and this benefits clock speed, price, and power consumption.

The PIC instruction set is suited to implementation of fast lookup tables in the program space. Such lookups take one instruction and two instruction cycles. Many functions can be modelled in this way. Optimization is facilitated by the relatively large program space of the PIC (e.g. 4096 × 14-bit words on the 16F690) and by the design of the instruction set, which allows for embedded constants. For example, a branch instruction's target may be indexed by W, and execute a "RETLW" which does as it is named - return with literal in W.

Interrupt latency is constant at three instruction cycles. External interrupts have to be synchronized with the four clock instruction cycle, otherwise there can be a one instruction cycle jitter. Internal interrupts are already synchronized. The constant interrupt latency allows PICs to achieve interrupt driven low jitter timing sequences. An example of this is a video sync pulse generator. This is no longer true in the newest PIC models, because they have a synchronous interrupt latency of three or four cycles.

5.2 Liquid Crystal Display (LCD)

LCD is a type of display used in digital watches and many portable computers. LCD displays utilize to sheets of polarizing material with a liquid crystal solution between them. An electric current passed through the liquid causes the crystals to align so that light cannot pass through them. LCD technology has advanced very rapidly since its initial inception over a decade ago for use in lap top computers. Technical achievements has resulted in brighter displace, higher resolutions, reduce response times and cheaper manufacturing process.

The liquid crystals can be manipulated through an applied electric voltage so that light is allowed to pass or is blocked. By carefully controlling where and what wavelength (colour) of light is allowed to pass, the LCD monitor is able to display images. A backlight provides LCD monitor’s brightness. Over the years many improvements have been made to LCD to help enhance resolution, image, sharpness and response times. One of the latest such advancement is applied to glass during acts as switch allowing control of light at the pixel level, greatly improving LCD’s ability to display small-sized fonts and image clearly.

Other advances have allowed LCD’s to greatly reduce liquid crystal cell response times. Response time is basically the amount of time it takes for a pixel to “change colours”, in reality response time is the amount of time it takes a liquid crystal cell to go from being active to inactive.

The ability to display numbers, characters and graphics. This is in contrast to LEDs, which are limited to numbers and a few characters. An intelligent LCD display of two lines, 20 characters per line that is interfaced to the pic16f72 microcontroller. Incorporation of a refreshing controller into the LCD, thereby relieving the CPU to keep displaying the data. Ease of programming for characters and graphics. Most of the LCD modules conform to a standard interface specification. A 14-pin access is provided having eight data lines, three control lines and three power lines. The connections are laid out in one of the two common configurations, either two rows of seven pins, or a single row of 14 pins. One of these pins is numbered on the LCD's printed circuit board (PCB), but if not, it is quite easy to locate pin1. Since this pin is connected to ground, it often has a thicker PCB track, connected to it, and it is generally connected to metal work at same point.

Vcc, Vss and Vee While Vcc and Vss provide +5V and ground respectively, Vee is used for controlling LCD contrast.RS Register Select there are two very important registers inside the LCD. The RS pin is used for their selection as follows. If RS=0, the instruction command code register is selected, allowing the user to send a command such as clear display, cursor at home, etc. If RS=1, the data register is selected, allowing the user to send data to be displayed on the LCD.R/W, read/write R/W input allows the user to write information to the LCD or read information from it. The pins Vcc, Vss, and Vee are essential for powering the LCD and adjusting its contrast. The **RS (Register Select)** pin determines whether commands or data are being sent to the LCD, while the **R/W (Read/Write)** pin controls the direction of data transfer. Together, these pins enable effective communication and control of the LCD display.

5.3 Relay Driver

Fig 5.1 Relay driver

A relay driver circuit is a circuit which can drive, or operate, a relay so that it can function appropriately in a circuit. The driven relay can then operate as a switch in the circuit which can open or close, according to the needs of the circuit and its operation. In this project, we will build a relay driver for both DC and AC relays. Since DC and AC voltages operate differently, to build relay drivers for them requires slightly different setup. We will also go over a generic relay driver which can operate from either AC or DC voltage and operate both AC and DC relays. All the circuits are relatively simple to understand. The contacts will suffer from these spikes as well. Surges in current that result from inductive effects can create very high voltage spikes (as high as 1000V) that can have nasty effects on neighbouring devices with in the circuits, such as switches and transistors getting zapped. , the circuit ensures that the relay can be activated with minimal control voltage, while also providing the necessary current to drive the relay coil.

5.4 Operation of DC Relay Driver

To drive a DC relay ,we will first go over how to build a relay driver circuit for relays which operate from DC power. To drive a DC relay, all we need is sufficient DC voltage which the relay is rated for and a Zener diode. All relays come with a voltage rating. This is called on a relay's datasheet its rated coil voltage. This is the voltage needed in order for the relay to be able to operate and be able to open or close its switch in a circuit. In order for a relay to function, it must receive this voltage at its coil terminals. Thus, if a relay has a rated voltage of 9VDC, it must receive 9 volts of DC voltage to operate. So the most important thing a DC relay needs is its rated DC voltage. If you don't know this, look up what relay you have and look up its datasheet and check for this specification.

And the reason why a diode is needed is usually because it functions to eliminate voltage spikes from a relay circuit as the relay opens and closes. The coil of a relay acts an inductor. Remember that inductors are basically coils of wires wrapped around a conductive core. This is what relay coils are as well. Therefore, they act as inductors. Inductors are electronic components that resist changes in current. Inductors do not like sudden changes in current. If the flow of current through a coil is suddenly interrupted.

Switch opening, the coil will respond by producing a sudden, very large voltage across its leads, causing a large surge of current through it. From a physics or physical perspective, this phenomenon is a result of a collapsing magnetic field within the coil as the current is terminated abruptly. Mathematically, this can be understood by noticing how a large change in current (dI/dt) affects the voltage across a coil.

The diode must be rated to handle currents equivalent to the maximum current that would have been flowing through the coil before the supply current was interrupted. Therefore, if the relay normally passes a certain amount of current through it during normal operation, the diode must be rated for a current rating above this value, as to not stop normal operation. Again, the DC relay must receive its rated voltage value in order to operate. The DC power source can be either batteries, wall wart power, or a DC power supply- any DC power source.

The Zener diode is placed reverse biased in parallel to the relay. A DC relay driver circuit allows the microcontroller to control high-power components like a relay using low-power signals. Since the PIC microcontroller cannot directly drive the relay coil, a transistor is used as a switch to energize the relay. A flyback diode is added across the relay coil to protect the transistor from voltage spikes. The base resistor ensures safe current flow into the transistor. This circuit is essential for isolating and safely disconnecting the load during fault conditions in the system. voltage spikes damaging to other electronic components in a circuit but they are also damaging to the relay's switch contacts.

The DC relay driver circuit effectively controls the operation of relays in the system, providing reliable switching between different components. By using transistors or MOSFETs as switching elements, the circuit ensures that the relay can be activated with minimal control voltage, while also providing the necessary current to drive the relay coil.

5.5 DC Relay Driver Circuit

The relay which we use in this case is rated for 9V. Therefore, a 9-volt DC voltage source feeds the resistor. To suppress transients that may be caused by the relay opening and closing, we place a Zener diode reverse biased in parallel with the relay. This will shunt all excess power to ground once it reaches a certain threshold. This is all that is needed to operate the relay. With sufficient power, the relay will now closed, driving the loads that are connected to its output.

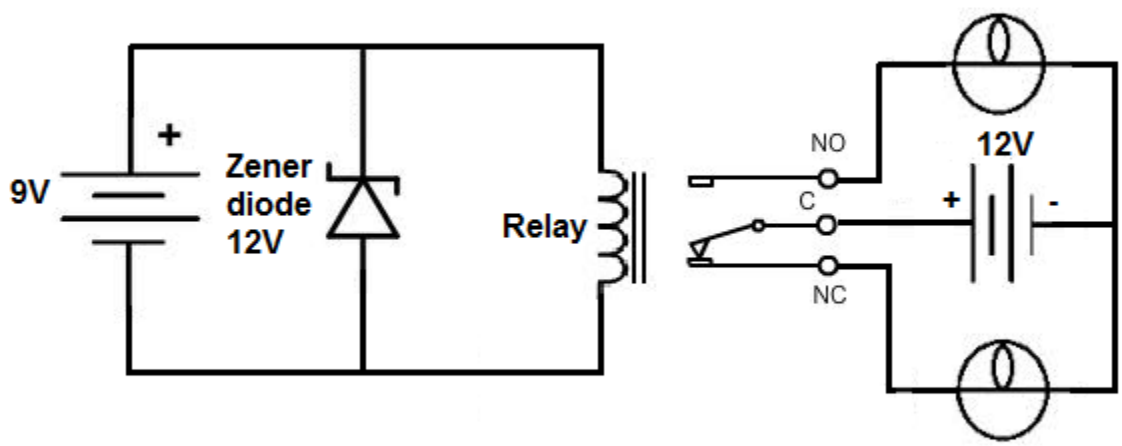

Fig 5.2 Schematic diagram of Relay driver

The last relay driver circuit we will show is one which can be driven by an arbitrary control voltage. This is a relay driver circuit which can be driven by either AC or DC input voltage. And unlike the other circuits, a specific voltage, such as the rated voltage values we used to drive the others, does not need to be used. Because this circuit contains a transistor, much less power needs to use on the input side to drive it.

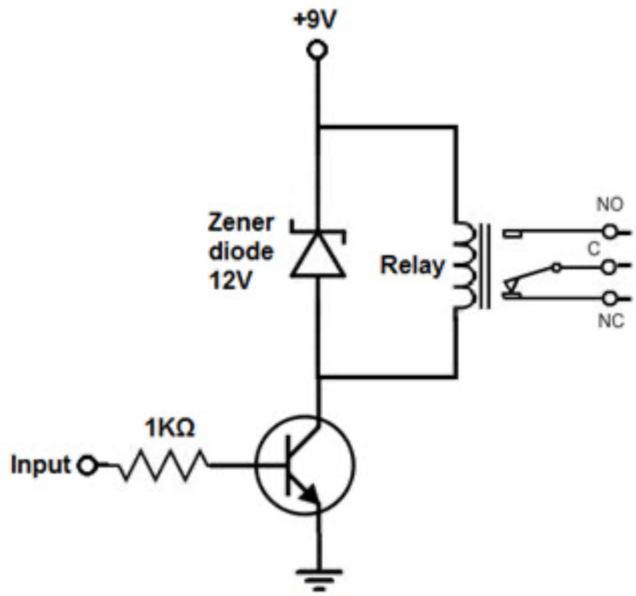

Fig 5.3 Circuit diagram of Generic Relay

In order to drive the relay, we use transistor and only less power can be possibly used to get the relay driven. Since, transistor is an amplifier so the base lead receives sufficient current to make more current flow from Emitter of Transistor to Collector. If the base once gets power that is sufficient, then the transistor conduct from Emitter to Collector and power the relay.

The Transistor's emitter-to-collector channel will be opened even though no input current or voltage is applied to Base lead of Transistor. Therefore, blocking current flows through relay coil. The emitter-to-collector channel will be opened and allows current to flow through relay's coil if enough current or voltage is applied as input to the base lead. AC or DC Current can be used to power the relay and circuit. Relays are electromagnetic devices which allow low-power circuit to switch a high current ON and OFF switching devices with the help of an armature that is moved by an electromagnet.

Driver Circuit is used to boost or amplify signals from micro-controllers to control power switches in semi-conductor devices. Driver circuits take functions that include isolating the control circuit and the power circuit, detecting malfunctions, storing and reporting failures to the control system, serving as a precaution against failure, analysing sensor signals and creating auxiliary voltages.

More interface options are available which includes popular ULN2003 driver. The ULN2003 has internal clamp diodes. While these work OK in non-critical applications and it leads to rise of glitches. The relay in this project acts as a critical safety switch that disconnects the battery or affected circuit when a fault or anomaly is detected. It is controlled by the microcontroller based on sensor data and LSTM model output. This prevents further damage by isolating the fault zone immediately. The relay ensures quick response to short circuits, overheating, or voltage imbalance.

5.6 REGULATOR 7805

Voltage sources in a circuit may have fluctuations resulting in not providing fixed voltage outputs. A voltage regulator IC maintains the output voltage at a constant value. 7805 IC, a member of 78xx series of fixed linear voltage regulators used to maintain such fluctuations, is a popular voltage regulator integrated circuit (IC). The xx in 78xx indicates the output voltage it provides. 7805 IC provides +5 volts regulated power supply with provisions to add a heat sink. All voltage sources cannot able to give fixed output due to fluctuations in the circuit. For getting constant and steady output, the voltage regulators are implemented.

The integrated circuits which are used for the regulation of voltage are termed as voltage regulator ICs. Here, we can discuss about IC 7805.The voltage regulator IC 7805 is actually a member of 78xx series of voltage regulator ICs. It is a fixed linear voltage regulator. The xx present in 78xx represents the value of the fixed output voltage that the particular IC provides. For 7805 IC, it is +5V DC regulated power supply. This regulator IC also adds a provision for a heat sink. The input voltage to this voltage regulator can be up to 35V, and this IC can give a constant 5V for any value of input less than or equal to 35V which is the threshold limit.

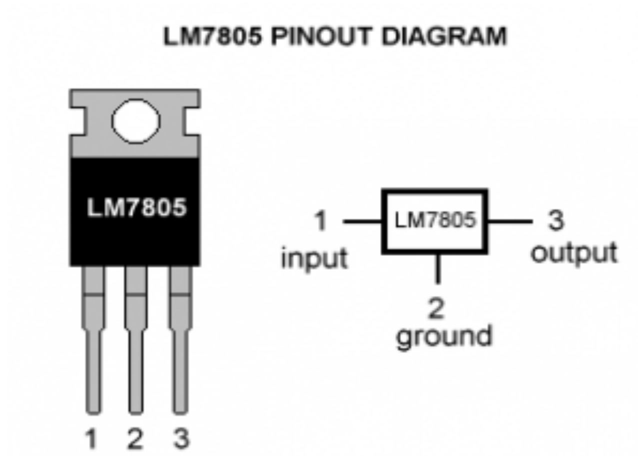

Fig 5.4 Regulator Pin Diagram

The purposes of coupling the components to the IC7805 are explained below. C1- It is the bypass capacitor, used to bypass very small extent spikes to the earth. C2 and C3- They are the filter capacitors. C2 is used to make the slow changes in the input voltage given to the circuit to the steady form. C3 is used to make the slow changes in the output voltage from the regulator in the circuit to the steady form. When the value of these capacitors increases, stabilization is enlarged.

Applications of Voltage Regulator 7805 IC Voltage and current regulators play a crucial role in powering electronic circuits with stable and reliable energy. A current regulator is essential in applications where a constant current is needed, such as in LED drivers and battery charging circuits. A regulated dual power supply is commonly used in operational amplifier circuits, analog signal processing, and other applications requiring both positive and negative voltage rails. These regulator circuits are frequently employed in building essential electronic devices such as phone chargers, UPS (Uninterruptible Power Supply) systems, and portable CD players, where maintaining a consistent voltage is critical to device functionality. A fixed output regulator provides a constant voltage level and is ideal for simple circuits requiring 5V or 12V power.

5.7 Power Supply and Battery management

The 3.7V Li-Ion battery pack plays a crucial role in powering the entire system for the Advanced Short Circuit Preventive Mechanism for Electric Vehicles. Lithium-Ion batteries are chosen due to their high energy density, compact size, long cycle life, and efficient power delivery, making them an ideal choice for applications like electric vehicles where space, weight, and performance are critical. A single 3.7V Li-Ion cell typically provides a nominal voltage of 3.7V, but in practice, several cells are connected in series and parallel configurations to achieve the desired system voltage and capacity. In a series connection, multiple cells are stacked together to increase the total voltage. For example, connecting three 3.7V cells in series results in a 11.1V pack (3.7V × 3).

Given their power and reliability, Li-Ion batteries must be carefully monitored to ensure safe operation. Overcharging, deep discharging, or overheating can damage the cells, which is why this system includes voltage sensors, current sensors, and temperature sensors. These sensors are integrated into the system to keep track of the battery's health in real-time. Voltage sensors detect any anomalies in voltage levels, while current sensors monitor any surge that could indicate a short circuit or overcurrent situation. Temperature sensors ensure that the cells do not overheat, preventing thermal runaway or potential fires.

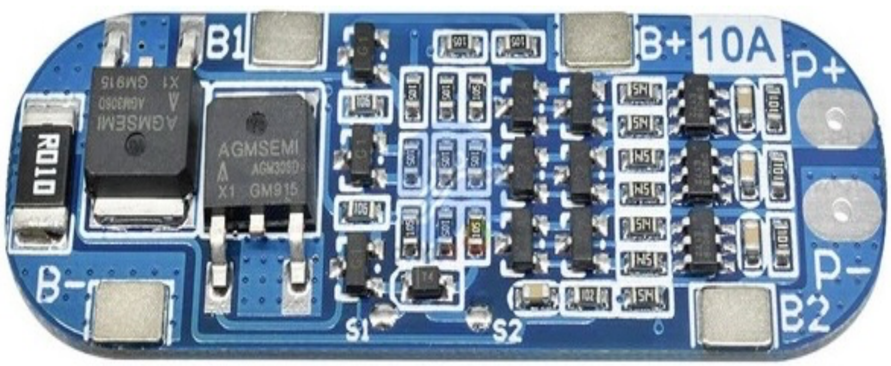

Fig 5.5 Battery Management System

This constant monitoring and protection system ensures that the 3.7V Li-Ion battery pack remains reliable and efficient throughout the operation of the electric vehicle. The integration of battery management with the fault detection and prevention mechanism allows the system to respond to abnormal conditions, activate safety protocols like disconnecting the battery pack, and help extend the longevity and safety of both the battery and the vehicle system as a whole.

Resistors play a vital role in controlling and limiting current flow within electrical circuits, ensuring that other components are protected from damage due to excessive current. In this project, resistors are strategically placed in various parts of the system to control the current flowing into sensitive components like the transistor in the relay driver circuit, voltage dividers for sensor circuits, and for current-limiting applications in the microcontroller’s input/output pins. Resistors are particularly important for protecting the base of the NPN transistor in the relay driver circuit.

By placing a base resistor, the current flowing into the transistor's base is regulated, ensuring that the transistor is not overwhelmed with excessive current, which could damage it. This helps to ensure reliable switching of the relay when triggered by the microcontroller’s signals. Similarly, pull-up or pull-down resistors are used to stabilize input pins and prevent floating voltage levels, which can cause erratic behaviours in digital circuits. In the voltage divider circuits for sensor calibration, resistors are used to scale the sensor output to a level that is within the readable range of the microcontroller's Analog-to-Digital converter (ADC). This ensures that accurate sensor readings are taken, helping to monitor parameters like voltage, current, and temperature effectively.

The choice of resistor values is crucial in achieving proper functionality of the circuit. Common resistors used in the project include values such as 1kΩ for base resistors in transistors, 10kΩ for pull-up/down configurations, and specific resistor values for sensor calibration that allow the system to read precise data and trigger safety mechanisms when necessary.

5.8 DC 5V Cooling Fan

The fan in this system operates on a 5V DC power supply, which aligns with the voltage levels available from the microcontroller and other low-power components in the system. This ensures compatibility with the existing electronics and prevents the need for additional voltage conversion, simplifying the design and reducing power loss.

Fig 5.6 DC Cooling Fan

The fan typically comes in a 40mm x 40mm x 10mm size, making it compact and suitable for tight spaces. Its small footprint allows it to be easily integrated into the system, particularly near the battery pack or relay circuit, without occupying much space. Despite its compact size, it delivers sufficient airflow for cooling. The fan operates at a low noise level of around 20 to 30 dB, ensuring that it does not contribute to unwanted noise during system operation. A quiet fan is especially important in applications where noise reduction is a priority.

CHAPTER 6

SOFTWARE REQUIREMENT

6.1 Embedded C

Embedded C is one of the most popular and most commonly used Programming Languages in the development of Embedded Systems. So, in this article, we will see some of the Basics of Embedded C Program and the Programming Structure of Embedded C. Embedded C is perhaps the most popular languages among Embedded Programmers for programming Embedded Systems. There are many popular programming languages like Assembly, BASIC, C++ etc. that are often used for developing Embedded Systems but Embedded C remains popular due to its efficiency, less development time and portability. Before digging in to the basics of Embedded C Program, we will first take a look at what an Embedded System is and the importance of Programming Language in Embedded Systems.

An Embedded System can be best described as a system which has both the hardware and software and is designed to do a specific task. A good example of an Embedded System, which many households have, is a Washing Machine. We use washing machines almost daily but wouldn't get the idea that it is an embedded system consisting of a Processor (and other hardware as well) and software. Embedded Systems can not only be stand-alone devices like Washing Machines but also be a part of a much larger system. An example of this is a Car. A modern-day Car has several individual embedded systems that perform their specific tasks with the aim of making the ride smooth, efficient, and safe.

Some of the embedded systems in a car are Anti-lock Braking System (ABS), Temperature Monitoring System, Automatic Climate Control, Tyre Pressure Monitoring System, Engine Oil Level Monitor, etc. In addition to automobiles, embedded systems are widely used in healthcare devices (like ECG machines and insulin pumps), industrial automation (robotic arms, sensors, controllers), consumer electronics (smartphones, smart TVs), and aviation systems. These systems are typically optimized for reliability, real-time performance, and energy efficiency. With the rise of IoT and AI, modern embedded systems are becoming more intelligent and interconnected, capable of predictive diagnostics and remote monitoring, making them an integral part of today's smart technology landscape.

6.2 Microcontroller Programming Using Embedded C

The software component of this project plays a critical role in managing data acquisition, processing sensor inputs, executing control logic, and implementing predictive analytics. A combination of embedded programming and machine learning tools has been used to ensure accurate anomaly detection, real-time fault response, and smooth system operation. The microcontroller in this project is programmed using Embedded C via the MPLAB X IDE with the XC8 compiler, which provides a flexible and efficient environment for low-level programming. This software setup allows precise control over hardware modules such as the relay, cooling fan, LCD display, and sensors (for temperature, current, and voltage). The embedded program handles tasks such as reading sensor data at regular intervals, comparing values with defined thresholds, and initiating protective actions when abnormalities are detected.

For the machine learning component, the LSTM (Long Short-Term Memory) algorithm is implemented using Python with libraries like TensorFlow and the LSTM model is trained on historical sensor data to learn normal operating patterns and identify deviations that indicate faults such as short circuits or imbalanced battery conditions. The trained model is then deployed on a connected edge device or computer for real-time inference. Additionally, NumPy, Matplotlib, and Pandas are used during the training phase for data handling and visualization.

Communication between the hardware and the ML model can be facilitated through serial communication using UART protocol, with data transferred via USB or wireless modules depending on the setup. The entire system is tested and debugged using simulation tools available within MPLAB and Python-based visualization for performance validation. In summary, the software requirements include both low-level embedded programming tools for real-time control and high-level machine learning frameworks for intelligent fault prediction. Together, these tools enable a seamless integration of hardware and intelligent software for a smart and proactive short circuit prevention system in electric vehicles.

6.3 Machine Learning Integration

One of the most advanced aspects of this project is the integration of Machine Learning (ML) to enhance the system's ability to detect anomalies and predict faults in real-time. The ML model used in this project is based on the Long Short-Term Memory (LSTM) algorithm, which belongs to the family of Recurrent Neural Networks (RNNs).

LSTM is specifically chosen due to its strong capability in learning from time-series data, such as the continuous stream of readings from current, voltage, and temperature sensors. The model is trained using a dataset collected from the hardware system under various operating conditions, including normal and fault scenarios. This dataset includes labelled data showing how parameters change over time during events like short circuits, overheating, or battery imbalance. During the training phase, tools like Python, TensorFlow, and NumPy are used to build and train the LSTM network. Visualization and analysis are carried out using Matplotlib and Pandas. Once trained, the LSTM model is capable of identifying subtle patterns that precede faults something that may be difficult to detect using traditional rule-based systems. The model continuously evaluates incoming sensor data and compares it with the learned patterns. If any abnormal behaviour is detected, such as a sudden spike in current or a drop in voltage, the model flags it as a potential fault.

Upon fault detection, the ML system sends a signal to the microcontroller to trigger a safety response such as disconnecting the relay or activating the cooling fan. This predictive and preventive approach not only protects the hardware components but also increases the overall reliability and safety of the system. Through the use of Machine Learning, the system transitions from being reactive to proactive, thereby adding a layer of intelligence that improves performance, safety, and efficiency.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Temperature Monitoring

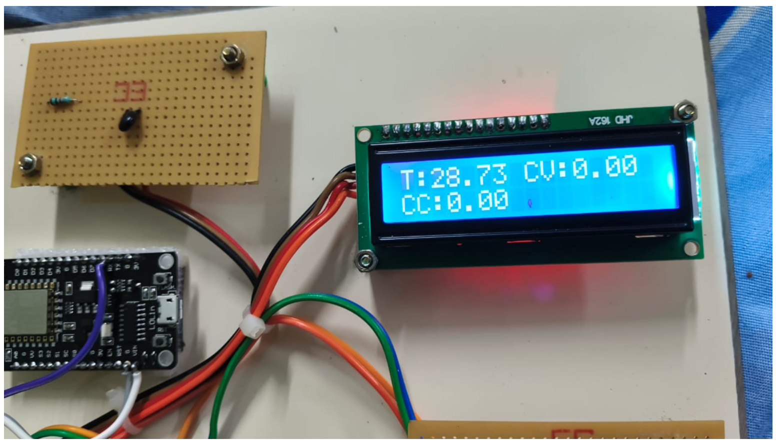

Fig.7.1 Tracking Temperature

Above result illustrates the system operating under normal monitoring conditions. The display shows a temperature reading of 28.80°C, with both charging current and voltage at zero, indicating that no charging activity is currently taking place. The state of charge is recorded at 98.34%, and the battery voltage is shown as 11.82V. These values suggest that the battery is nearly fully charged, which explains the absence of active charging. The stable readings on the LCD validate the system's ability to accurately monitor the battery's condition in real time. This confirms that the sensors and the microcontroller are functioning as intended, delivering consistent and reliable data.

7.2 Voltage and Current Monitoring

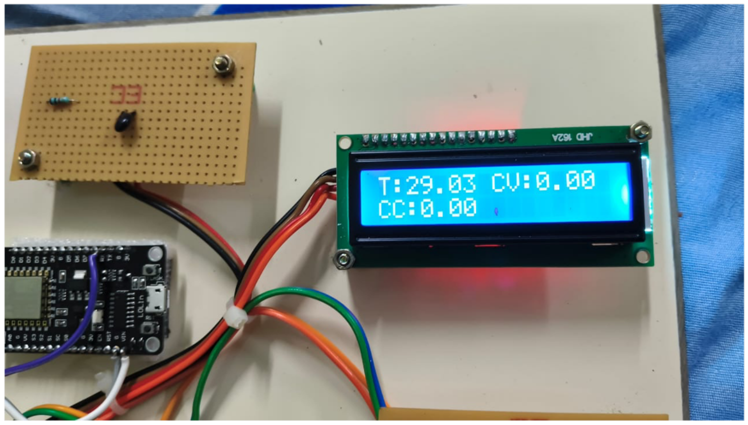

Fig.7.2 Monitoring Voltage and Current

Above results displays the system in an idle state, where all parameters are initialized but no charging activity is occurring. The LCD screen shows a temperature reading of 29.03°C, with both the charging current (CC) and charging voltage (CV) reading 0.00. This condition typically represents a state where the system is powered on and sensors are functioning, but the battery is either not connected or not in need of charging. The consistency and accuracy of the displayed values confirm that the microcontroller is correctly interpreting and presenting sensor data. Additionally, the neatly arranged components and clean wiring reinforce the quality of the hardware setup. This image showcases the baseline condition of the monitoring system, demonstrating that it is ready to respond as soon as battery activity is detected. The idle state also allows for system diagnostics without risking battery wear. It reflects a controlled environment for testing signal stability and sensor calibration. Such stability at rest validates the system's readiness for real-time operation.

7.3 Overcharge Detection

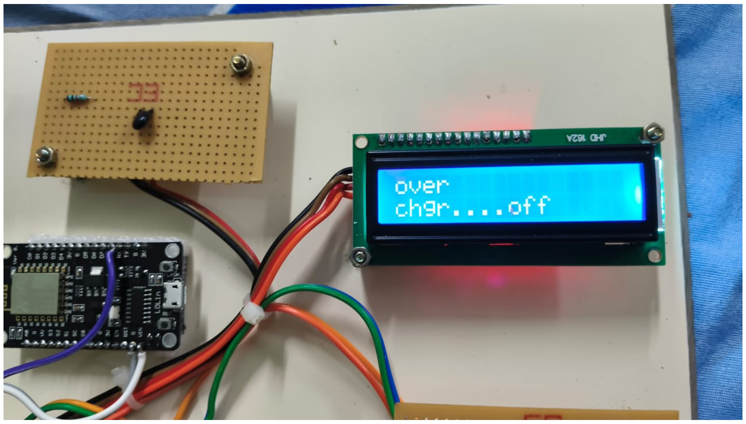

Fig.7.3 Overcharging Detection

Above results captures the system's response when an overcharging condition is detected. The display shows a warning message indicating that charging has been turned off due to an overcharge. This demonstrates the system's protective mechanism in action, where the battery charging process is automatically disabled once a critical threshold is reached. Such a response is vital to prevent battery damage, overheating, or potential safety hazards. This feature emphasizes the practical application of automated control in ensuring battery safety and supports the system’s reliability in managing critical battery events without user intervention. This not only enhances operational safety but also extends the lifespan of the battery. The alert mechanism is clearly visible and easy to interpret, reducing the chances of human error. Overall, this functionality reflects a thoughtful integration of safety protocols into the system’s core logic.

7.4 State of Charge and Battery Voltage

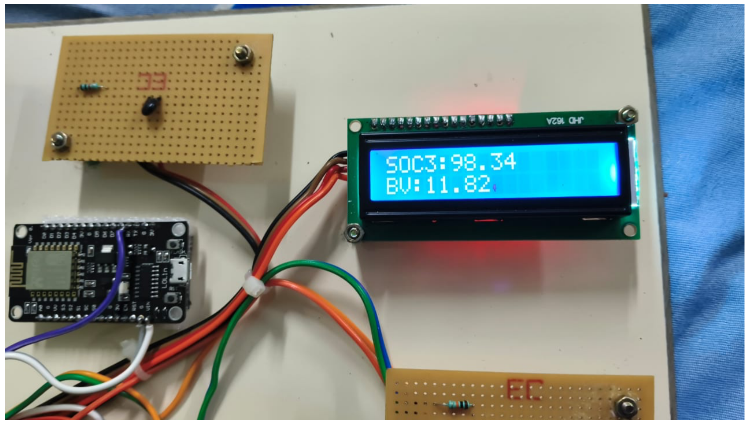

Fig.7.4 Monitoring State of Charge

Above results continues to reflect the system's real-time monitoring capability under stable battery conditions. The temperature is shown as 28.80°C, with charging current and voltage remaining at zero. The state of charge is maintained at 98.34%, and the battery voltage is once again displayed as 11.82V. The repetition of these values across multiple images further supports the claim that the system consistently delivers accurate data over time. This scenario demonstrates that the monitoring system is capable of sustained performance and that the displayed values align with the actual battery condition, making it a dependable tool for battery health management.It also highlights the effectiveness of the sensor calibration and software logic. The system’s non-intrusive observation mode allows for passive diagnostics without interrupting normal operation. This functionality is crucial for real-world applications where reliability and continuous monitoring are essential.

CHAPTER 8

CONCLUSION AND FUTURE WORK

This research is innovative because it takes a proactive approach to electric car short circuit prevention by combining self-recovery capabilities, predictive fault detection, and real- time monitoring. In contrast to conventional reactive systems, this model analyses sensor data and predicts any errors before they happen by using sophisticated machine learning methods like Recurrent Neural Networks (RNNs) and Gradient Boosted Trees. High-speed semiconductor switches minimize system damage and downtime by ensuring microsecond-level fault isolation. Furthermore, the self-diagnostic module eliminates the need for user intervention by automatically restoring system functionality.

A new standard for safety, dependability, and efficiency in EV power systems is set by this creative combination of automated recovery, rapid reaction mechanisms, and predictive analytics. This strategy lays the groundwork for upcoming developments in EV safety technology in addition to addressing the shortcomings of conventional fault avoidance systems. The system guarantees adaptability to a variety of operating settings by utilizing Recurrent Neural Networks for temporal pattern analysis and Gradient Boosted Trees for accurate anomaly identification. By combining these state-of-the-art methods, the system becomes more resilient and scalable, which makes it a strong answer to the changing needs of electric mobility.

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***Abstract*— In order to improve safety, dependability, and system performance, the proposed project introduces a cutting-edge short circuit preventative mechanism made especially for electric vehicles (EVs). In EV power systems, short circuits are a serious problem since they can result in serious harm, component failures, and safety risks. Traditional protection techniques, including circuit breakers and fuses, are frequently slow, reactive, and unable to handle the complex and dynamic working circumstances of contemporary EVs to proactively prevent short circuits, this research makes use of a combination of high-speed sensing, predictive fault detection algorithms, and intelligent control tactics. The system uses machine learning techniques to predict and alleviate issues before they increase by continuously monitoring critical metrics including current, voltage, and temperature variations. To guarantee quick reaction and strong protection, sophisticated parts are integrated, such as semiconductor switches and thermal management subsystems. In order to minimize downtime and maintenance expenses, the suggested mechanism also includes self-diagnostic capabilities, which allow for continuous monitoring and automatic recovery in the event of temporary defects. The system's ability to detect and prevent short circuits is demonstrated by simulation and experimental findings, which beat conventional protection techniques in terms of speed, accuracy, and dependability This invention contributes to safer, more effective, and sustainable EV systems by offering a comprehensive solution to one of the most important problems in electric mobility. By laying the groundwork for next-generation electrical safety solutions, the project hopes to increase EV adoption and confidence. In addition to tackling important issues in electric mobility, this innovation helps create EV systems that are safer, more effective, and more sustainable. It creates a solid basis for broad EV adoption by boosting user confidence and system resilience, hastening the shift to a cleaner and more sustainable future. Along with its technical advantages, the system's self-diagnostic features allow for automatic recovery from transient malfunctions, constant monitoring, and a reduction in maintenance needs, all of which drastically save operating expenses. Its scalable and modular architecture ensures industry-wide usability by allowing it to be integrated into a variety of EV platforms, from private automobiles to commercial fleets.**

I. Introduction

The growing popularity of electric vehicles (EVs) has brought attention to how important it is to have dependable and effective electrical systems in order to guarantee both safety and peak performance. Short circuits are one of the many issues that EV power systems encounter, and they are especially dangerous since they can result in serious harm, such as system failures, thermal runaway, and even user safety risks. As a result, a top focus in automobile research and development is now creating sophisticated preventive mechanisms to identify, lessen, and control short circuits in EV power systems. Traditional short circuit protection techniques frequently involve fuses or circuit breakers, which have intrinsic drawbacks such slower response times, no real-time monitoring, and restricted reusability, despite being useful in some situations. These difficulties call for the creation of increasingly complex systems that can detect faults proactively, react quickly, and recover on their own. This study suggests a sophisticated short circuit prevention system made especially for electric cars that incorporates state-of-the-art technology including active power flow management, machine learning- based predictive diagnostics, and real-time fault detection algorithms. The risk of short circuits is further increased by the dynamic and constantly shifting working circumstances of EV power systems, which include fluctuating loads, high currents, and extremely high temperatures. Advanced preventive systems that can adjust in real-time to guarantee system stability and safety are required to meet these difficulties. In order to provide thorough and dependable protection for the entire system, effective short circuit avoidance systems must take these complexity into consideration by including features like adaptive control techniques, fault-tolerant designs, and improved thermal management.

The proposed mechanism not only enhances the safety and reliability of EV power systems but also contributes to extending the lifespan of critical components and reducing downtime. By integrating advanced analytics and control strategies, this solution aligns with the broader goal of fostering sustainable and safe transportation technologies. The effectiveness of the mechanism is validated through simulation and experimental results, demonstrating significant improvements in fault prevention and system robustness compared to traditional approaches.

Background of the Invention

Electric vehicles (EVs) are rapidly gaining popularity due to their potential to reduce greenhouse gas emissions and dependence on fossil fuels But there are still significant issues with EV power systems' dependability and security. Short circuits, which can have disastrous consequences like component failure, thermal runaway, fire dangers, and even the risk of human death, are a major worry in these systems. A number of things can cause short circuits, such as poor insulation, conductive objects, bad wiring, or unforeseen environmental circumstances like moisture or extremely high or low temperatures.. To address short circuits, conventional protective devices like fuses and mechanical circuit breakers are frequently employed. Although these solutions offer rudimentary security, they frequently fail to meet the particular difficulties presented by EV situations. For example, EVs are susceptible to fast cycles of charging and discharging and operate under a variety of load circumstances, which can put stress on the electrical system. Furthermore, traditional systems are reactive rather than proactive in their approach since they are unable to monitor, predict, and avoid failures in real-time [6]. The development of more effective and efficient protective methods has been made possible in recent years by developments in semiconductor technology, sensor systems, and intelligent control units. High-speed switches, advanced algorithms, and machine learning tools enable real-time monitoring and predictive fault detection, paving the way for active and preemptive short circuit prevention.

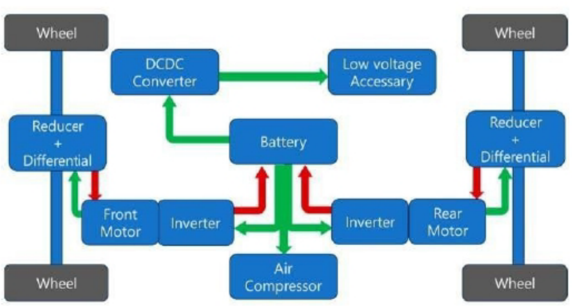

Fig. 1. Work Flow Diagram

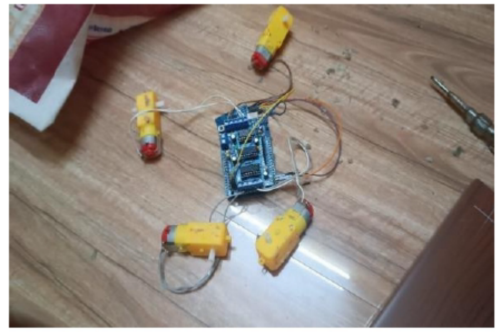

Fig. 2. Initial Fault Detection System

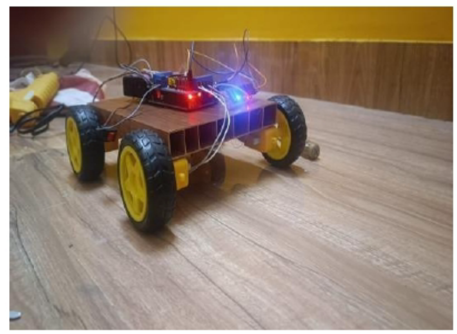

Fig. 3. E V System performance

II. Objective of the Invention

The invention's main goal is to create a cutting-edge short circuit prevention system for electric vehicles (EVs) that will increase system durability, safety, and dependability. By employing real-time monitoring, predictive diagnostics, and intelligent control strategies, the mechanism seeks to proactively identify and mitigate electrical defects, lowering risks, decreasing downtime, and enhancing the overall functionality of EV power systems. The goal of this idea is to get beyond the drawbacks of traditional protection techniques like circuit breakers and fuses, which are frequently reactive, slower, and unable to handle the dynamic operating circumstances of EVs. By utilizing cutting-edge technology including high-speed sensors, predictive defect detection algorithms, and intelligent control units, the suggested approach emphasizes proactive prevention [6]. In order to enable the system to identify, anticipate, and react to possible short circuits with remarkable speed and accuracy, it seeks to continuously monitor vital factors such as current, voltage, and thermal fluctuations. Another key objective is to integrate self-diagnostic and recovery capabilities, allowing the system to autonomously manage transient faults and minimize downtime. The invention also emphasizes scalability and cost-effectiveness, ensuring its adaptability across a wide range of EV models and operating conditions. Ultimately, this invention aspires to enhance the overall safety and reliability of EVs, fostering consumer confidence and supporting the transition to sustainable electric mobility by mitigating one of the most pressing electrical challenges in the automotive industry.

III. Summary of the Invention

This invention introduces an advanced short circuit preventive mechanism specifically designed for electric vehicles (EVs) to address critical safety and reliability concerns. Unlike traditional protection methods such as fuses and circuit breakers, this system offers a proactive approach by combining real-time monitoring, predictive analytics, and intelligent control strategies. The mechanism uses high-speed sensors to continuously monitor parameters like current, voltage, and temperature, enabling rapid detection of anomalies indicative of a potential short circuit. Machine learning algorithms are employed to analyze these parameters, allowing the system to predict and mitigate faults before they occur. In addition, advanced semiconductor switches ensure rapid response and precise isolation of faulty circuits, minimizing damage and preventing cascading failures To enhance operational reliability, the system is equipped with self-diagnostic and recovery features, allowing it to autonomously manage transient faults and resume normal operation without human intervention. This reduces downtime and maintenance costs while extending the lifespan of critical EV components. Designed for scalability, the invention can be integrated into a wide range of EV platforms and customized to suit varying operational conditions. By ensuring robust protection, improved efficiency, and reduced risks, this innovative mechanism significantly enhances the safety and dependability of EV power systems.

Fig. 4. Model of EV

IV. Detailed Description of the Invention

The increasing use of electric vehicles (EVs) has made it more difficult to guarantee the dependability and safety of their electrical systems. Short circuits present serious concerns, including as system failures, thermal runaway, and fire threats, among other difficulties. Although they can be somewhat effective, traditional solutions like fuses and mechanical circuit breakers are reactive and cannot keep up with the changing needs of contemporary EV systems.

By implementing a sophisticated short circuit prevention method, our invention overcomes these constraints. It improves system longevity and safety by proactively detecting and mitigating defects through the use of intelligent control techniques, real-time monitoring, and predictive analytics.

A. System Architecture

The suggested short circuit preventative mechanism's system design consists of multiple interconnected parts that cooperate to provide electric vehicles with complete protection. High-speed sensors give the central control unit real-time data by continuously monitoring important parameters including temperature, voltage, and current. In order to anticipate and categorize such issues before they arise, the control unit uses sophisticated machine learning algorithms to process this data. Semiconductor switches minimize disturbance and avoid damage by isolating malfunctioning circuits in microseconds upon detecting anomalies.

B. Semiconductor switching system

An essential part of the suggested short circuit prevention mechanism, which is intended to offer quick and dependable fault separation in electric vehicles (EVs), is the semiconductor switching system. Semiconductor switches function electrically as opposed to mechanically, allowing for quicker reaction times and increased robustness. The system uses solid-state components that can withstand the high currents and voltages seen in EV power systems, such as Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) and Insulated Gate Bipolar Transistors (IGBTs). The control unit alerts the semiconductor switches to isolate the impacted area of the circuit when it detects a possible short circuit. Sensitive components are much less likely to sustain damage thanks to this isolation, which happens in microseconds. These switches' arc-free operation ensures longevity and dependability even under heavy use by removing the wear and tear that comes with mechanical systems. Additionally, the energy-efficient design of the semiconductor switches minimizes power losses when in regular use. They are perfect for integration into a variety of EV architectures, from heavy-duty electric trucks to passenger cars, due to their small size and great scalability. Furthermore, the system may operate in both directions, which makes it appropriate for EV battery systems' charging and discharging procedures. The semiconductor switches are made to identify and react to malfunctions in less than ten microseconds. This quick reaction time is essential for stopping the spread of faults and reducing damage to parts like motors, batteries, and inverters. Power flows in both directions in EV systems: when the battery is charged and when energy is being released to the motor. Bidirectional currents can be handled by semiconductor switches with ease, guaranteeing reliable functioning in a variety of scenarios. When in use, mechanical switches frequently generate arcs, which can degrade materials and pose safety hazards. Since semiconductor switches are entirely electronic, these problems are resolved, guaranteeing more robust and secure operation. Because of its low on-state resistance, the switches produce less heat and lose less energy when in use. This enhances the range of the EV and improves overall energy efficiency and efficiency of energy use.

C. System speed

Operational speed is significantly improved by implementing advanced high-speed sensors that provide real-time data on critical parameters like current, voltage, and temperature. These inputs are processed by a central control unit equipped with machine learning algorithms, ensuring rapid fault detection and immediate response through high-precision semiconductor switches.

D. Downtime Minimization

Downtime in electric vehicle systems is greatly reduced by combining self-recovery mechanisms with real-time system diagnostics. These systems keep an eye on the functioning of important parts, like battery subsystems and power electronics, in order to spot irregularities early. With the use of sophisticated machine learning algorithms, predictive maintenance techniques examine both historical and current data to predict probable faults. This proactive approach lessens the need for physical interventions while also assisting in identifying and correcting problems before they become more serious. The overall dependability and effectiveness of the vehicle are improved by making sure that temporary issues are automatically fixed and that vital systems are quickly returned to regular functioning. By avoiding serious damage or prolonged outages, these features also increase component lifespan and save maintenance expenses. Predictive maintenance methods use both past and present data to forecast likely issues, like battery pack cell imbalances or connector deterioration. Because minor disorders are handled before they become more serious, this proactive approach lessens the need for physical treatments. For example, by identifying and isolating malfunctioning battery cells, Tesla's real-time diagnostic system can preserve overall performance and plan maintenance for off-peak hours. The vehicle's efficiency and dependability are greatly increased by making sure that transient problems are automatically fixed and vital systems are promptly restored. Additionally, these features prolong component lifespan and lower maintenance costs by preventing serious damage or prolonged downtime,

E. Managing Costs

Saving people and fleets thousands of rupees a year. By employing modular components that grow with system requirements, lowering initial investment, and permitting upgrades only when required, cost-effectiveness is guaranteed. Early defect detection via predictive diagnostics reduces maintenance costs, which are normally between ₹8,000 and ₹40,000 per component, and prevents serious failures. Energy-efficient designs, including low-resistance MOSFETs, save ₹5,000 to ₹10,000 per car yearly by reducing operating energy expenses by 10-15%. By decreasing downtime, automated self-recovery systems save ₹3,000 to ₹12,000 a year in manual intervention expenses. Together, these actions minimize costs while preserving system dependability.

F. Selection of Algorithm

To guarantee efficient defect detection and prediction, the algorithm chosen for this project places a strong emphasis on accuracy, speed, and adaptability. Gradient Boosted Trees (GBT) are used because of its excellent anomaly detection accuracy and capacity to describe intricate, non-linear relationships in sensor data. GBT is a dependable option for real-time system monitoring since it is especially well-suited to managing a variety of data inputs, including measurements of temperature, voltage, and current. In addition, temporal dependencies that are crucial for detecting and forecasting developing defects in electric car systems are captured by Recurrent Neural Networks (RNNs), which analyze sequential data. By combining the strengths of RNN in processing temporal patterns with GBT in static data analysis, this synergy creates a strong foundation for proactive fault control. The integrated strategy reduces system downtime, guarantees quick fault isolation.

G. Prediction of short circuit

High-speed sensors are used in the suggested system to continuously monitor important parameters including temperature, voltage, and current. For instance, the central control unit (CCU) receives the data if the current increases from 50 A to 60 A in milliseconds. This is recognized as an abnormality outside of the typical operating range by the CCU, which is driven by machine learning algorithms. Then, using past patterns—like recurrent current spikes or a sharp rise in temperature from 40°C to 70°C—it forecasts a possible short circuit. Semiconductor switches are predicted to isolate the problematic circuit in less than 10 microseconds [2]. This proactive measure guarantees safe operation, safeguards components, and stops cascading failures. The occurrence is recorded by the system, which also sends out a diagnostic notice for upkeep.

H. Fault detection system

Prediction and mitigation of this kind improve enhance safety, reduce downtime, and prolong system life in electric vehicles. By merging the outputs of several decision trees to increase prediction accuracy, the suggested method improves fault identification using sophisticated techniques such as Gradient Boosted Trees for accurate anomaly detection. Recurrent neural networks (RNNs) are perfect for tracking changing EV situations and forecasting possible faults over time because they can analyze temporal patterns and sequences in sensor data. Fault profiling is further strengthened by multi-parameter analysis that makes use of temperature, voltage, and current data. Robust monitoring is ensured by hybrid sensor networks that incorporate impedance and Hall effect sensors. While redundant sensors continue to function in the event of a failure, edge computing allows real-time fault isolation within 5 microseconds. After a fault, self-healing algorithms dynamically reorganize the system to continue functioning. Proactive repairs are made possible by real-time alerts and predictive maintenance, which identify patterns. These improvements guarantee quick and accurate fault identification, increasing EV safety and reliability. Using a hybrid sensor network that consists of both voltage and current sensors, the system detects any sudden voltage drop that occurs during fast charging in an EV. While Recurrent Neural Networks (RNNs) examine the sequence of sensor data to detect any impending problems, the Gradient Boosted Trees (GBT) technique precisely forecasts the fault's nature [9]. Using self-healing algorithms built into the control unit, edge computing with FPGA-based processing units isolates the problematic component in a matter of microseconds and reconfigures the system. Through the infotainment system, the driver receives a real-time alert about a possible issue, allowing them to schedule maintenance or repairs before any serious harm is done and guaranteeing continuous operation.

I. Thermal Management System

In electric vehicles (EVs), the thermal management subsystem is essential to preserving the effectiveness, security, and durability of the short circuit prevention system. Excessive heat is produced in the system's components, especially in the power electronics and semiconductor switches, under high-load circumstances such short circuits or abrupt power fluctuations. The thermal management subsystem maintains ideal temperature control to avoid overheating, which can result in system failure, thermal deterioration, or even fires. Active and passive cooling techniques are combined to accomplish this. High-power components, such semiconductor switches and battery packs, are usually directly cooled by active cooling techniques like liquid cooling systems or forced air circulation [2]. Heat sinks and thermal pads are examples of passive cooling techniques that remove excess heat without the need for extra electricity or moving elements. Real-time temperature sensors are integrated into the thermal management subsystem to keep an eye on vital components. When the system detects temperatures above safe thresholds, cooling mechanisms are activated. Furthermore, sophisticated heat spreaders are used to disperse heat evenly and avoid localized hotspots that might harm delicate parts. The thermal management subsystem keeps the system's operating temperature consistently within a safe range, which helps to increase efficiency and performance while also preventing the EV from overheating in fault situations. This increases the overall safety of the vehicle's power system, decreases heat-related energy losses, and extends the life of electrical components.

V. Components Used

A. High Speed Sensors

High-speed sensors are essential to the operation of electric vehicles' (EVs') sophisticated short circuit prevention system. Throughout the EV's power system, these sensors continuously and in real time measure vital electrical parameters like current, voltage, temperature, and impedance. Their main responsibility is to identify any departures from typical operating circumstances that can point to the beginning of a short circuit or other issues. The system can act quickly to stop harm because these sensors are made to have incredibly quick response times, usually less than one millisecond. For instance, current sensors—which are frequently based on Hall effect technology—are able to measure the system's current flow precisely and identify anomalous decreases or surges that could be signs of short circuits. These high-speed sensors continuously send data to the central control unit, which processes it to determine the short circuit risk and initiate the appropriate preventive actions. Utilizing these cutting-edge sensors improves the overall safety and dependability of the car's electrical system by enabling early fault detection and quick isolation of problematic areas [5]. The flow of current through different parts, such as the battery, inverter, and electric motor, is measured using current sensors, such as Hall effect sensors. They are able to promptly detect anomalous current decreases or surges, which frequently indicate the presence of an overcurrent or short circuit. The voltage levels across important parts of the EV's electrical system are continuously measured by these sensors. A short circuit or unstable power supply are two examples of imminent faults that are frequently indicated by voltage variations or departures from predetermined thresholds. When voltage anomalies are detected quickly, the system can take preventative action, including rerouting power or turning off problematic areas.

B. Central Control Unit (CCU)

The central control unit (CCU), which coordinates the entire fault detection, diagnosis, and protection process, is the brains behind the electric vehicle's (EV) short circuit prevention system. By analyzing real-time data from high-speed sensors and controlling the system's reaction to possible malfunctions, it serves as the decision-making hub and guarantees the dependability and safety of the car's electrical system. A high-performance microprocessor or microcontroller, which can carry out intricate algorithms with little delay, lies at the heart of the CCU. These algorithms, which are frequently powered by artificial intelligence or machine learning, examine sensor data to find anomalous patterns that could signal the start of a short circuit or other electrical problems [8]. These patterns include current surges, voltage drops, temperature spikes, and impedance changes. By comparing real-time sensor data with predetermined safety criteria, the CCU continuously assesses the system's health and makes sure that any deviations from standard operating conditions are promptly detected and fixed. When a failure is identified, the CCU immediately initiates the required precautionary measures, such as turning on cooling systems to avoid overheating, rerouting power, or isolating the problematic circuit by turning on semiconductor switches.

C. Semiconductor Switches

Electric vehicles' (EVs') short circuit prevention system relies heavily on semiconductor switches, which isolate malfunctioning circuits quickly, effectively, and dependably. Semiconductor switches use solid-state technology, which gives them a number of important advantages over conventional mechanical circuit breakers or fuses, including quicker response times, increased longevity, and more accuracy. Advanced semiconductor materials such as Metal- Oxide-Semiconductor Field-Effect Transistors (MOSFETs) and Insulated Gate Bipolar Transistors (IGBTs), which can manage high current and voltage levels while reducing power losses, are commonly used to make these switches. Within microseconds of the system detecting a malfunction, like a short circuit, the semiconductor switches isolate the faulty circuit and stop it from spreading to other areas of the vehicle's power supply. This quick reaction greatly lowers the possibility of damaging vital parts like batteries, motors, and inverters, which can be expensive to replace or repair.

D. Power Distribution

In electric vehicles (EVs), the power distribution system is a crucial subsystem that controls the effective transfer of electrical energy from the battery to different parts including the motor, inverters, and auxiliary systems. In order to ensure smooth vehicle operation and safety, particularly in the event of a breakdown, this system is essential in making sure that power is distributed in an efficient and controlled manner. In order to direct and control electrical power throughout the vehicle's architecture, the power distribution system is made up of a network of high-capacity connections, power distribution units (PDUs), and protective devices. Central energy storage is provided by the battery pack, and the power distribution system makes sure that energy is allocated to the engine for propulsion, the onboard chargers for recharging, and the auxiliary loads such as lights, infotainment, and climate control. The power distribution system's capacity to monitor and control voltage and current levels is a crucial component that helps avoid overvoltage or overcurrent scenarios that could cause damage or malfunctions. In order to stop additional damage, the system includes a number of safety devices, including fuses, circuit breakers, and semiconductor switches, which immediately isolate impacted sections when they detect fault conditions like overloads or short circuits. When a short circuit or other defect occurs, the central control unit (CCU) and semiconductor switches cooperate with the power distribution system to quickly cut off the problematic area, redirect power to areas that are not affected, and keep the system running continuously. Furthermore, the power distribution system is often equipped with high-speed communication lines to relay real-time information to the CCU, which processes the data and triggers corrective actions, such as activating cooling systems or adjusting power flow. The system is designed for scalability, allowing for integration with larger, more powerful systems in electric trucks, buses, or commercial vehicles.

E. Protection System

To protect the electrical architecture of electric vehicles (EVs) from defects such short circuits, overvoltage, overcurrent, and thermal overload, the protection system is an essential part that ensures the vehicle's safety as well as the integrity of its parts. This system is in charge of continuously checking the power system's condition, identifying anomalies in real time, and acting quickly to fix them in order to protect vital parts including the battery pack, motor, inverters, and wiring. Usually, the protection system consists of a mix of software and hardware elements that cooperate to identify, isolate, and lessen fault circumstances as they appear. Overcurrent protection, which makes use of fuses, circuit breakers, and solid-state semiconductors, is one of the main elements of the protective system. switches to interrupt the power flow when excessive current is detected. These gadgets are precisely adjusted to react to abrupt spikes in current, which may be a sign of short circuits or other hazardous electrical issues. Overvoltage protection, which guarantees that the voltage levels across different components stay within safe working limits, is another essential component.

To stop damage from electrical surges or spikes, the protection system will cut off or restrict power to the impacted components if it senses that the voltage has risen above the predetermined level. Since overheating can result in thermal runaway or even fires, thermal protection is also a crucial component of the protection system [4]. Temperature levels are regularly monitored by temperature sensors mounted on high-heat parts including motors, semiconductor switches, and batteries. To avoid thermal damage, the system turns on cooling mechanisms or turns off particular components if a temperature increase is detected that exceeds safe bounds.

F. Self-Diagnostic and Feedback Module

A vital component of electric vehicles' (EVs') short circuit prevention system, the Self-Diagnostic and Feedback Module improves the vehicle's overall performance, safety, and dependability. This module provides real-time examination of the vehicle's components, including the battery, power distribution system, sensors, semiconductor switches, and control units, and continuously checks the condition and functionality of the complete electrical system. The module can identify anomalies, malfunctions, or performance deterioration long before they become serious failures by utilizing sophisticated diagnostic algorithms. It continuously assesses the condition of the car's electrical subsystems using information obtained from a variety of high-speed sensors, control units, and power management systems [1]. The Self-Diagnostic and Feedback Module initiates an instantaneous reaction upon detecting a fault, be it an anomalous spike in current, an increase in temperature, a variation in voltage, or the failure of a particular component. This response may involve the activation of the protection system, the isolation of malfunctioning components, or the redirection of power to unaffected areas. The module documents the fault's kind, location, and severity in addition to taking prompt corrective action. This information can be utilized for analysis and optimization in the future. Through the use of machine learning and predictive analytics, this feedback loop is crucial for improving the system's capacity to anticipate possible errors. This helps to anticipate wear and tear or other performance problems before they result in system breakdowns. Service staff can monitor and troubleshoot remotely because to the feedback module's ability to interact with external diagnostic tools. By ensuring that problems are found and fixed before they worsen, this remote diagnostic capability minimizes downtime and lowers maintenance expenses. By offering insightful information on the vehicle's energy usage, operational effectiveness, and general health, the module also contributes significantly to system optimization [7]. This information may be utilized to improve the performance of the power system and other crucial parts. The driver can also receive warnings from the Self-Diagnostic and Feedback Module, which gives them up-to-date information on the electrical system's condition. The driver can be empowered to take preventative action by receiving notifications of possible problems, performance deterioration, or maintenance requirements. Through constant system performance evaluation, data recording, and actionable insights, the Self-Diagnostic and Feedback Module not only guarantees optimal vehicle operation but also helps to continuously enhance the vehicle's overall efficiency and safety. Real-time data sharing and cloud-based analytics for improved system insights are made possible by the Self-Diagnostic and Feedback Module's smooth integration with IoT platforms [2]. Additionally, it facilitates over- the-air upgrades, which enable ongoing enhancements to diagnostic algorithms and the addition of new defect detection features. Furthermore, by learning from past data, the module facilitates adaptive reactions, guaranteeing improved precision in identifying and addressing complicated errors. The longevity of EV components is greatly increased by this proactive approach, which also increases user confidence in the dependability of the vehicle.

G. Communication System

In order to facilitate effective operation, safety, and performance management, the communication system in electric vehicles (EVs) is a crucial piece of infrastructure that guarantees smooth data flow between several subsystems, components, and external systems. The Central Control Unit (CCU), sensors, safety features, power distribution units, and other important subsystems inside the car can all communicate in real time thanks to this technology. It is the foundation of the car's electrical and diagnostic system, making sure that every part works together and reacts fast to any changes in the system's condition. Strong and secure protocols that provide dependable, fast data transfer with low latency, such Ethernet, wireless communication, and the Controller Area Network (CAN) bus, are usually the foundation of a communication system. By transmitting essential data, including sensor readings, fault detection signals, system status updates, and diagnostic information, these protocols make sure that every subsystem is informed of the general health of the vehicle in real time In order to identify possible problems, the CCU, for example, continuously gets input from high-speed sensors that monitor variables including current, voltage, temperature, and impedance. When a fault is detected, the communication system promptly notifies the appropriate subsystems, including the protection system, so that they can take prompt action to isolate the problematic component and stop damage [10]. Furthermore, the communication system makes sure that every part—including the self-diagnostic module, thermal management systems, and semiconductor switches—can effectively communicate with one another.

H. Fuse and Circuit Breaker

In the electrical architecture of electric vehicles (EVs), fuses and circuit breakers are vital protective elements that protect the vehicle's power supply from overcurrent situations that could cause serious harm or safety risks. These gadgets are made to identify excessive current flow, which can be brought on by overloads, short circuits, or broken parts, and cut off the faulty circuit to stop additional harm. Fuses are straightforward but efficient safety devices made especially to guard against the dangers of overcurrent situations in electrical circuits. When the current above a preset threshold, a fuse's tiny metal strip or wire melts, thus stopping the flow of electricity [11]. A fuse is a one-time- use device since it must be replaced after blowing. Fuses are commonly employed in lower-power applications that require a prompt and decisive reaction to overcurrent situations. Because they shield sensitive parts from harmful currents, they are particularly useful for safeguarding smaller electrical subsystems, sensors, and battery packs. In contrast, circuit breakers provide a more advanced, resettable solution using a thermal or magnetic mechanism that detects the increase in current and disconnects the circuit to prevent damage to the system. Circuit breakers are similar to fuses in that they are made to trip and disconnect a circuit when excessive current is detected. but they can be reset manually or automatically after they have been tripped. This makes them more appropriate for higher-power applications where overcurrent conditions may occur more frequently but need to be addressed without replacing the component each time.

VI. Current Limiting Devices

In order to prevent excessive current flow—which can result from faults like short circuits, overloads, or malfunctioning components—from damaging the electrical architecture of electric vehicles (EVs), current limiting devices are crucial parts of their electrical systems. These gadgets are made to regulate the current that passes through a circuit so that it stays within safe operating bounds. Current limiting devices aid in preventing overheating, component deterioration, and possible system failure by actively regulating the current under fault conditions. These factors are crucial for preserving the vehicle's dependability, performance, and safety. There are several types of current limiting devices, such as resistive components, active current limiters, and passive current limiters. Because of their intrinsic resistance, passive current limiters like fuses and resistors are made to lower the current to a safe level. For instance, fuses function by adding a regulated quantity of resistance to the circuit [3]. The fuse efficiently disconnects the circuit and stops more damage when the current reaches a predetermined level. In some applications, resistors can also be employed as current limiters, acting as a buffer to absorb excess current and shield the remainder of the system from damaging power surges. All things considered, current limiting devices are essential to preserving the longevity, safety, and effectiveness of electric car electrical systems because they guarantee that they function within safe bounds and reduce the possibility of major component damage.

VII. Design

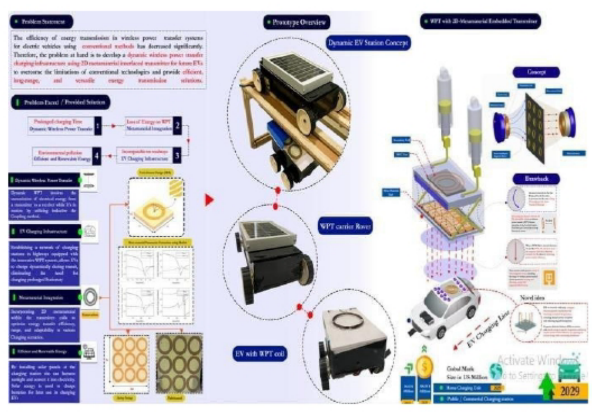

Fig. 5. System Architecture

S(MVA)	U1（kV）	U2（kV）	θ(°)	S_f(MVA)	h(%)
200	11.10	10.56	07438	198.1	-0.95
210	11.08	10.56	0.6927	208.0	-0.95
220	11.05	10.56	0.6476	217.9	-0.95
230	11.03	10.56	0.6076	227.8	-0.96
240	11.01	10.55	0.5719	237.7	-0.96
250	10.99	10.55	0.5399	247.6	-0.96
260	10.97	10.55	0.5111	257.5	-0.96
270	10.95	10.55	0.4849	267.4	-0.96
280	10.94	10.55	0.4611	277.3	-0.96
290	10.92	10.55	0.4395	287.2	-0.97
300	10.91	10.55	0.4196	297.1	-0.97

Fig. 6. Sample Data

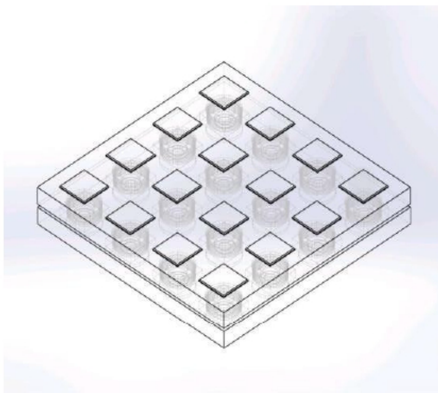

Fig. 7. Transparent design view of car top

In order to guarantee the electrical safety, stability, and dependability of the vehicle, the short circuit preventative mechanism for electric vehicles (EVs) is designed as a highly sensitive and efficient system that incorporates several subsystems. The design's major goal is to preserve the vehicle's peak performance while offering smooth protection against issues like short circuits, over currents, and overvoltage situations. Advanced sensors, semiconductor switches, protection devices, communication systems, and current limiting devices are just a few of the hardware and software elements that are incorporated into the design. These elements all cooperate to identify and address issues before they have a chance to cause serious harm. The Central Control Unit (CCU), which is at the center of the design, gets data in real time from a variety of sensors that track important parameters including temperature, voltage, and current. The CCU can detect possible system hazards such an overcurrent or a short circuit using this information. The circuit breakers and semiconductor switches are triggered to isolate the problem and disconnect the impacted circuit when a fault is identified. Even in the case of a malfunction, the power system will continue to function within safe bounds thanks to the assistance of current limiting devices, which help regulate the flow of electricity.

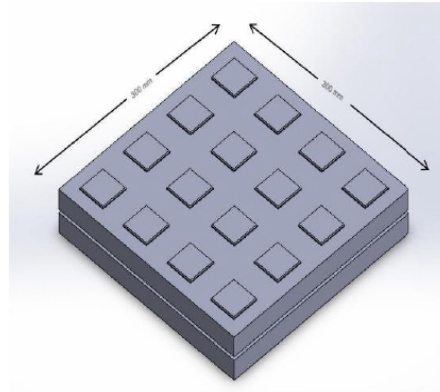

Fig. 8. Complete design view of top of car

VIII. System comparison

A. Limitations of Existing

Traditional EV fault detection systems are slow, with response times in milliseconds, inadequate for dynamic scenarios like thermal runaway events in lithium-ion batteries. They lack predictive diagnostics and real-time monitoring, leading to costly battery replacements (~₹4,000,000) and increased downtime due to reliance on mechanical fuses. These limitations hinder effective fault isolation and fail to prevent thermal events, raising critical safety concerns. Advanced solutions with faster response times and robust diagnostics are essential to address these challenges.

B. Proposed model comparison

Important technical domains, the suggested approach shows notable improvements over conventional fault detection methods. While the suggested approach uses sophisticated semiconductor switches to provide microsecond-level responses (<5 μs), typical systems rely on mechanical components, such as fuses and circuit breakers, with reaction times in the range of 10–100 milliseconds. Another crucial area is fault detection methods, where conventional systems use threshold-based reactive detection that is devoid of predictive capabilities. On the other hand, proactive anomaly identification is made possible by the suggested model's real-time analysis of multi-parameter data using machine learning methods. The suggested method uses solid-state semiconductor switches, such as MOSFETs and IGBTs, for quick, accurate, and arc-free fault isolation, whereas conventional systems rely on mechanical components, which are slower and more prone to wear [3]. Traditional thermal management systems usually use passive techniques, like thermal pads and heat sinks, which react slowly to overheating. For efficient and dynamic thermal control, the suggested approach incorporates active techniques including liquid cooling and real-time temperature monitoring. Furthermore, traditional systems have limited scalability and adaptability, which limits their application to particular configurations. The suggested system is more flexible and effective due to its modular design.

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IX. Conclusion

This research is innovative because it takes a proactive approach to electric car short circuit prevention by combining self-recovery capabilities, predictive fault detection, and real-time monitoring. In contrast to conventional reactive systems, this model analyses sensor data and predicts any errors before they happen by using sophisticated machine learning methods like Recurrent Neural Networks (RNNs) and Gradient Boosted Trees. High-speed semiconductor switches minimize system damage and downtime by ensuring microsecond-level fault isolation. Furthermore, the self-diagnostic module eliminates the need for user intervention by automatically restoring system functionality. A new standard for safety, dependability, and efficiency in EV power systems is set by this creative combination of automated recovery, rapid reaction mechanisms, and predictive analytics. This strategy lays the groundwork for upcoming developments in EV safety technology in addition to addressing the shortcomings of conventional fault avoidance systems. The system guarantees adaptability to a variety of operating settings by utilizing Recurrent Neural Networks for temporal pattern analysis and Gradient Boosted Trees for accurate anomaly identification. By combining these state-of-the-art methods, the system becomes more resilient and scalable, which makes it a strong answer to the changing needs of electric mobility. In the end, this research opens the door for EV systems that are more dependable, effective, and sustainable, which boosts consumer confidence and encourages a wider uptake of electric vehicles.

X. References

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